



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

**OBJECT ORIENTED PROGRAMING
PRACTICES
(UNC 302)**

Assignment-1
Group: 2023

Submitted By:-

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LAB -1

Topic Covered - cin/cout basics, structures in C, structures in C++, Classes and objects

Programming Question

1. Write a program (WAP) to display "Hello World" on console display.
Code:

```
#include<iostream>
using namespace std;
int main(){
    cout<<"Hello World";
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab>
b_assignment1 }
Hello World
```

2. WAP to that reads two values from keyboard and displays the larger value on the screen.
Code:

```
#include<iostream>
using namespace std;
int main(){
    int a,b;
    cin>>a>>b;
    int x=a>b?a:b;
    cout<<x<<endl;
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab>
b_assignment1 }
3 4
4
```

3. WAP to read three values a, b, c from keyboard and display the value of $x = a/b - c$.

Code:

```
#include<iostream>
using namespace std;
int main(){
    int a,b,c;
    cin>>a>>b>>c;
    double x=a/b-c;
    cout<<x<<endl;
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab>
b_assignment1 }
11
2
3
2
```

4. WAP to implement the following control characters:

- '*n*' is for new line, or you can use *endl* – *cout<<endl<<"message";*
- '*t*' is for tab
- '*a*' is an alarm sound
- '*r*' is carriage return to go to the beginning of the current line

Code:

```
#include<iostream>
using namespace std;
int main(){
    cout<<"My name is Abhinav.\n I live in jaipur"<<endl;
    cout<<"I am \t 20 years old"<<endl;
    cout<<"\a"<<endl;
    cout<<"\r end"<<endl;
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab>
b_assignment1 }
My name is Abhinav.
I live in jaipur
I am      20 years old

end
```

5. Implement *namespace* in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.

Code:

```
#include<iostream>
using namespace std;
namespace first{
    int x=1;
    void display(){
        cout<<"The value of x is: "<<x<<endl;
    }
}
namespace second{
    int x=2;
    void display(){
        cout<<"The value of x is: "<<x<<endl;
    }
}
int main(){
    cout<<first::x<<endl;
    cout<<second::x<<endl;
    first::display();
    second::display();
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab>
b_assignment1 }
1
2
The value of x is: 1
The value of x is: 2
```

6. Write a program to define a structure *Student* that contains 3 variables and 2 functions:

- *Variables: (a)*

name

(b) roll

(c) marks

- *Functions:*

(1) setStudentData()

(2) getStudentData()

Use loops to input and output data for 3 student object entries.

Code:

```
#include<iostream>
using namespace std;
struct Student{
    char name[20];
    int roll;
    float marks;
    void setStudentdata(){
        cout<<"Enter student name: ";
        cin>>name;
        cout<<"Enter student Roll No.: ";
        cin>>roll;
        cout<<"Enter student Marks: ";
        cin>>marks;
    }
    void getStudentdata(){
        cout<<"Student name: "<<name;
        cout<<" Student Roll No.: "<<roll;
        cout<<" Student Marks: "<<marks<<endl;
    }
};
int main(){
    Student s[3];
    for(int i=0;i<3;i++)
        s[i].setStudentdata();
    for(int i=0;i<3;i++)
        s[i].getStudentdata();
    return 0;
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab> cd "c:\Users\ab
b_assignment1 }
Enter student name: Abhinav
Enter student Roll No.: 194
Enter student Marks: 65
Enter student name: Aaveg
Enter student Roll No.: 203
Enter student Marks: 70
Enter student name: Ronak
Enter student Roll No.: 140
Enter student Marks: 23
Student name: AbhinavStudent Roll No.: 194Student Marks: 65
Student name: AavegStudent Roll No.: 203Student Marks: 70
Student name: RonakStudent Roll No.: 140Student Marks: 23
```

7. In the previous program make the data private, and functions public.
Code:

```
#include<iostream>
using namespace std;
class Student{
private:
    char name[20];
    int roll;
    float marks;
public:
    void setStudentdata(){
        cout<<"Enter student name: ";
        cin>>name;
        cout<<"Enter student Roll No.: ";
        cin>>roll;
        cout<<"Enter student Marks: ";
        cin>>marks;
    }
    void getStudentdata(){
        cout<<"Student name: "<<name;
        cout<<" Student Roll No.: "<<roll;
        cout<<" Student Marks: "<<marks<<endl;
    }
};
int main(){
    Student s[3];
    for(int i=0;i<3;i++)
        s[i].setStudentdata();
    for(int i=0;i<3;i++)
        s[i].getStudentdata();
    return 0;
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab> cd "c:\Users\abhin\OneDrive\Desktop\OOPS Lab"
b_assignment1 }
Enter student name: Abhinav
Enter student Roll No.: 194
Enter student Marks: 60
Enter student name: Aaveg
Enter student Roll No.: 123
Enter student Marks: 231
Enter student name: Rohan
Enter student Roll No.: 123
Enter student Marks: 89
Student name: Abhinav Student Roll No.: 194 Student Marks: 60
Student name: Aaveg Student Roll No.: 123 Student Marks: 231
Student name: Rohan Student Roll No.: 123 Student Marks: 89
```

8. Convert program #4 from struct to class.

Code:

```
#include<iostream>
using namespace std;
class Student{
private:
    char name[20];
    int roll;
    float marks;
public:
    void setStudentdata(){
        cout<<"Enter student name: ";
        cin>>name;
        cout<<"Enter student Roll No.: ";
        cin>>roll;
        cout<<"Enter student Marks: ";
        cin>>marks;
    }
    void getStudentdata(){
        cout<<"Student name: "<<name;
        cout<<" Student Roll No.: "<<roll;
        cout<<" Student Marks: "<<marks<<endl;
    }
};
int main(){
    Student s[3];
    for(int i=0;i<3;i++)
        s[i].setStudentdata();
    for(int i=0;i<3;i++)
        s[i].getStudentdata();
    return 0;
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab> cd "c:\Users\abhin\OneDrive\Desktop\OOPS Lab"
b_assignment1 }
Enter student name: Abhinav
Enter student Roll No.: 194
Enter student Marks: 60
Enter student name: Aaveg
Enter student Roll No.: 123
Enter student Marks: 231
Enter student name: Rohan
Enter student Roll No.: 123
Enter student Marks: 89
Student name: Abhinav Student Roll No.: 194 Student Marks: 60
Student name: Aaveg Student Roll No.: 123 Student Marks: 231
Student name: Rohan Student Roll No.: 123 Student Marks: 89
```

9. Discuss the differences of structures in C and structures in C++ during the lab with other students or your instructor.
10. WAP to read an integer value from the keyboard and display on screen “WELL DONE” that many times.

Code:

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cout<<"How many times you want to print well done?\n";
    cin>>n;
    for(int i=1;i<=n;i++)
        cout<<"Well Done"<<endl;
    return 0;
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab>
b_assignment1 }
How many times you want to print well done?
4
Well Done
Well Done
Well Done
Well Done
```

11. WAP to utilize assignment += and -= type operators.

Code:

```
#include<iostream>
using namespace std;
int main(){
    int x,y;
    cout<<"Enter value of x: ";
    cin>>x;
    cout<<"Enter value of y: ";
    cin>>y;
    x+=y;
    cout<<"After x+=y, x= "<<x<<endl;
    x-=y;
    cout<<"After x-=y;x= "<<x<<endl;
    return 0;
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab>
b_assignment1 }
Enter value of x: 25
Enter value of y: 50
After x+=y, x= 75
After x-=y;x= 25
```

12.WAP to swap two numbers using bitwise operators.

Code:

```
#include<iostream>
using namespace std;
int main(){
    int a,b;
    cout<<"Enter value for a:";
    cin>>a;
    cout<<"Enter value for b:";
    cin>>b;

    a = a ^ b;
    b = a ^ b;
    a = a ^ b;

    cout<<"After swapping:"<<endl;
    cout<<"a="<<a<<endl;
    cout<<"b="<<b<<endl;
    return 0;
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab>
b_assignment1 }
Enter value for b:32
After swapping:
a=32
b=23
```

13. Write a program to solve the following problem:

A library charges a fine for every book return late. For first 5 days the fine is 50 paise, for 6–10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled.

WAP to accept number of days the member is late to return the book and display the fine or appropriate message.

Code:

```
#include<iostream>
using namespace std;
int main(){
    int days;
    cout<<"How many days was he delay?";
    cin>>days;

    if(days <= 5){
        cout<<"Fine = 0.50 rupees";
    }
    else if(days <= 10){
        cout<<"Fine = 1.00 rupees";
    }
    else if(days <= 30){
        cout<<"Fine = 5.00 rupees";
    }
    else{
        cout<<"Membership cancelled";
    }

    return 0;
}
```

Output:

```
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab>
b_assignment1 }
Fine = 0.50 rupees
PS C:\Users\abhin\OneDrive\Desktop\OOPS Lab>
b_assignment1 }
b_assignment1 }
How many days was he delay?40
Membership cancelled
```